

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a0514da36a5a7738a.jsonl`  
- SHA-256: `30b2cc3caf0c3c774250d53c98a07574f9e74de4e58b4929d7480f5077f89ffa`  
- Source modified: `2026-05-31T04:25:00+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:24:12.659Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's model WEIGHTS are NOT permissively licensed: they are governed by the 'AI PUBS OPEN RAIL-M LICENSE (MODIFIED)', Version 0.1 (March 2, 2023) ? a restricted (OpenRAIL-M family) license, not MIT/BSD/Apache/ISC/HPND. This makes the weights non-compliant with muneem's hard constraint #3 (permissive weights required)."

****Source:**** https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
****Supporting quote:**** "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)"

Version 0.1, March 2, 2023 (Modified)
<http://licenses.ai/>"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:24:16.294Z)

I'll verify this claim about Surya's model weights license. Let me check the primary source and look for contradicting evidence.

3. assistant (2026-05-31T04:24:19.840Z)

I have the tools loaded. Let me fetch the primary source and search for contradicting evidence in parallel.

4. user (2026-05-31T04:24:25.266Z)

License Analysis

****Exact License Header (verbatim):****

"AI PUBS OPEN RAIL-M LICENSE (MODIFIED) Version 0.1, March 2, 2023 (Modified)"

****License Family:****

This is an OpenRAIL-M family license, not a permissive license like MIT/BSD/Apache. OpenRAIL-M stands for "Open Responsible AI Licenses for Models."

****Usage Restrictions Present:****

Yes, there are significant restrictions. Attachment A outlines "USE RESTRICTIONS" including:

- ****Legal restrictions:**** Prohibits violations of applicable laws and third-party intellectual property infringement
- ****Commercial restrictions:**** Includes revenue/funding thresholds?entities exceeding \$5,000,000 in gross revenue or total equity/debt funding cannot use it commercially without a separate commercial license
- ****Competitive restrictions:**** Prevents use by entities offering competing products or services to the licensor

The license explicitly requires these use-based restrictions be passed to all downstream users and incorporated into derivative works.

5. user (2026-05-31T04:24:27.600Z)

Web search results for query: "Surya OCR datalab model license OpenRAIL-M commercial use"

Links: [{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"}, {"title":"surya/MODEL_LICENSE at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"}, {"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"}, {"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-ocr/"}, {"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"}, {"title":"surya/LICENSE at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/LICENSE"}, {"title":"chandra/MODEL_LICENSE at master · datalab-to/chandra","url":"https://github.com/datalab-to/chandra/blob/master/MODEL_LICENSE"}, {"title":"Datalab - Intelligent Document Processing: IDP-Software","url":"https://idp-software.com/vendors/datalab/"}, {"title":"Datalab develops small AI models to power 'document intelligence' in the enterprise","url":"https://www.forkable.io/p/datalab-develops-small-ai-models"}, {"title":"Surya OCR toolkit - Datalab","url":"https://api.datalab.to/surya"}]

Based on the search results, here's what I found about the Surya OCR model's license and commercial use terms:

License Overview

The Surya code is licensed under Apache 2.0. However, the model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

Commercial Use

The license has restrictions on commercial use. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue). For broader commercial licensing of the model weights, visit our pricing page here.

More specifically, larger companies need a commercial license from Datalab to remove GPL code requirements and access enterprise support.

Commercial and broader use licenses may be available from Licensor at the following URL:
<https://www.datalab.to/>

In summary, Surya's OpenRAIL-M license permits free use for research, personal projects, and startups under \$5M in funding/revenue, but organizations beyond that threshold must obtain a paid commercial license from Datalab.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:24:30.117Z)

Both the primary source and the search confirm the claim. Let me verify the exact license name on the README and the HuggingFace weights page to confirm there's no permissive dual-license option for the weights themselves.

7. user (2026-05-31T04:24:35.531Z)

Surya License Summary

(a) CODE License

****Apache 2.0**** ? The code uses a permissive open-source license.

(b) MODEL WEIGHTS License

The model weights use a ****modified AI Pubs Open Rail-M license****. According to the README:

```
> "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)."
```

For broader commercial use, the project directs users to their pricing page.

Permissive Options for Model Weights?

****No.**** There is no MIT/BSD/Apache permissive option stated for the model weights themselves. The weights are governed by the Open Rail-M variant, which has restrictions tied to use case (research/personal/startup limitations). Commercial licensing requires a separate agreement through their pricing page.

8. user (2026-05-31T04:24:37.141Z)

License for Surya OCR 2 Model Weights

According to the web page content, the model weights use a ****modified AI Pubs Open Rail-M license****.

The specific statement from the page is:

```
> "The Surya code is licensed under Apache 2.0. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue). For broader commercial licensing of the model weights, visit our pricing page [here](https://www.datalab.to/pricing?utm_source=gh-surya)."
```

Additionally, the page metadata indicates:

- ****License: openrail****

Key Points:

- ****Code****: Apache 2.0
- ****Model Weights****: Modified AI Pubs Open Rail-M
- ****Permitted Uses****: Research, personal use, and startups under \$5M in funding/revenue
- ****Commercial Use****: Requires licensing through their pricing page

9. assistant (2026-05-31T04:24:47.682Z)

The claim is fully corroborated by multiple primary sources. Let me apply the adversarial checklist:

****Checklist verdict:****

1. ****Supported by quote?**** Yes. The MODEL_LICENSE file header reads verbatim "AI PUBS OPEN RAIL-M LICENSE (MODIFIED) Version 0.1, March 2, 2023 (Modified)" ? exactly matching the claim. This is unambiguously an OpenRAIL-M family license, not MIT/BSD/Apache/ISC/HPND.
2. ****Contradicting evidence?**** None found. Every source (MODEL_LICENSE file, GitHub README, HuggingFace metadata tagged `license: openrail`, datalab.to) agrees the **weights** are OpenRAIL-M while only the **code** is Apache-2.0. The claim correctly distinguishes weights from code.
3. ****Source quality?**** Primary source (the license file itself in the official `datalab-to/surya` repo) plus corroborating official README and HF model card. Sufficient for the claim's strength.
4. ****Outdated?**** The repo is the current `datalab-to/surya` (the maintained namespace as of 2025-2026; the sibling `chandra` repo uses the same MODEL_LICENSE). No newer permissive relicensing found. Claim is current.
5. ****Marketing/speculation?**** No ? it's a verbatim license-file quote.

****One minor nuance (does not refute):**** The claim describes it as a "restricted (OpenRAIL-M family) license." Sources reveal it is specifically a **modified** OpenRAIL-M with a \$5M revenue/funding threshold (free below, commercial license required above), plus competitive-use restrictions. This makes the license **more** restrictive than vanilla OpenRAIL-M, strengthening rather than weakening the claim's core assertion that the weights are non-permissive and non-compliant with hard constraint #3. The constraint requires MIT/BSD/Apache/ISC/HPND; OpenRAIL-M (modified or not) is none of these.

The claim is well-supported, current, and backed by primary sources.

10. user (2026-05-31T04:24:55.438Z)

Structured output provided successfully

11. assistant (2026-05-31T04:25:00.172Z)

I've completed the verification. The claim is NOT refuted (verified as true).

The claim that Surya's model weights are governed by the "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" ? a restricted OpenRAIL-M family license rather than a permissive one ? is fully supported by primary sources and current. The license file header matches the quote verbatim, and three independent sources (GitHub README, HuggingFace model card tagged `license: openrail`, and datalab.to) corroborate the code/weights license split. If anything, the actual terms (\$5M revenue/funding cap plus competitive-use restrictions) are more restrictive than the claim states, strengthening its conclusion that the weights violate muneem's hard constraint #3.